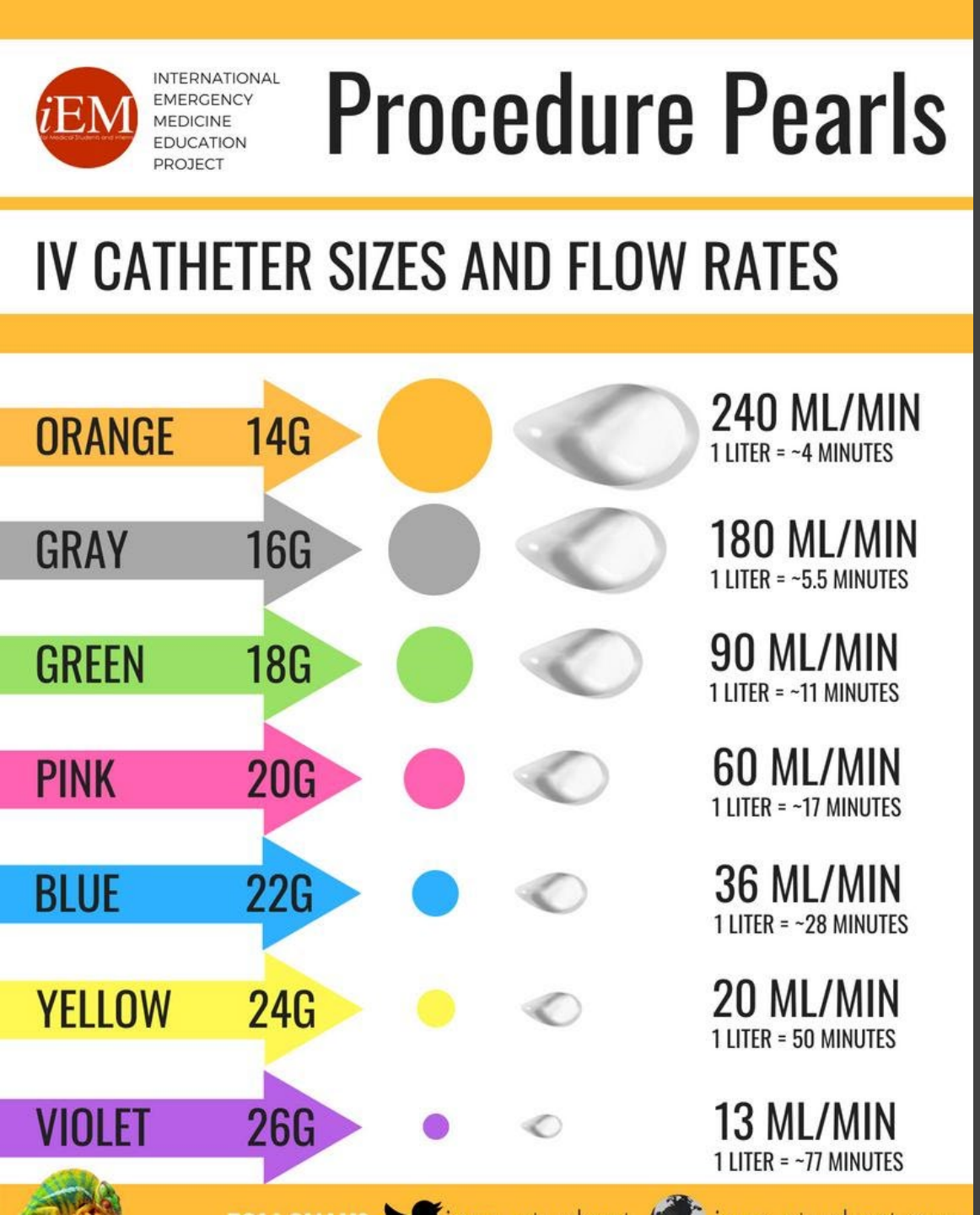
Medication Administration

Outline

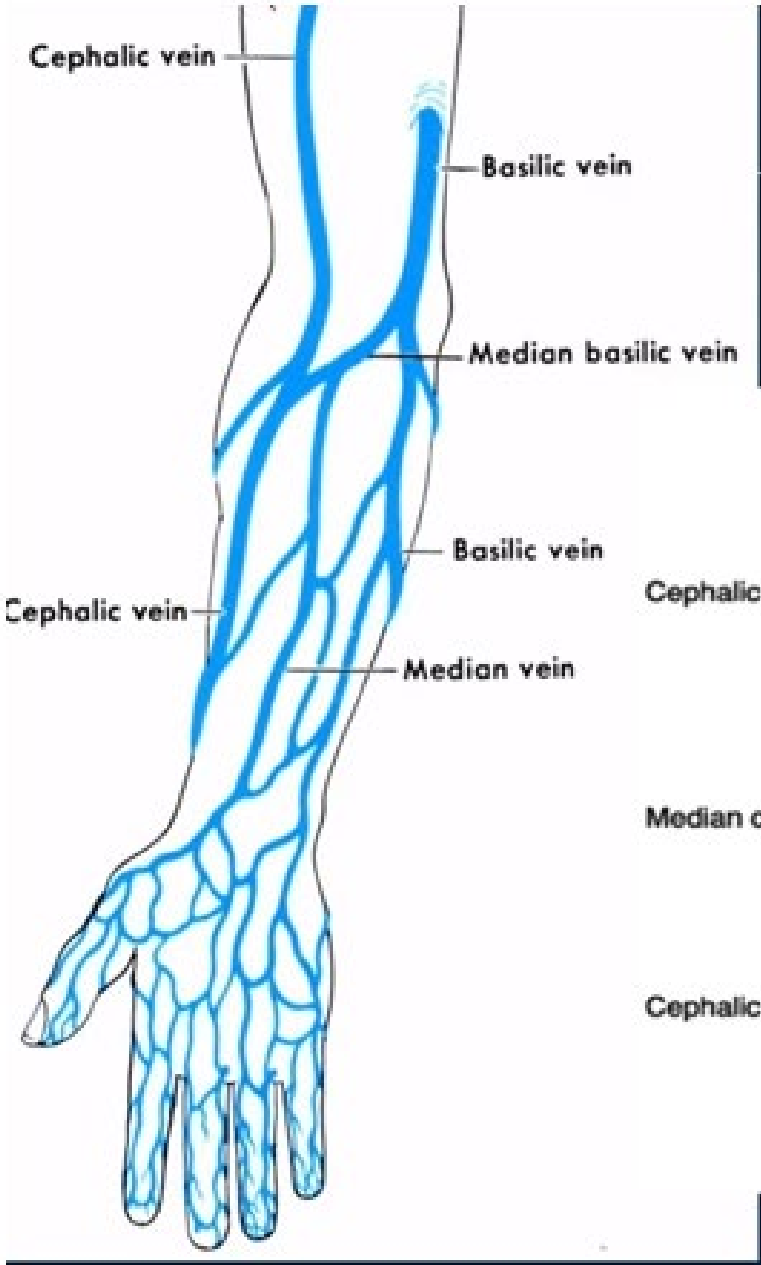
- Vascular Access
- Drip Rate Calculations
- Pressure Bags
- B-Braun
- Medication Infusions
- Antibiotic Reconstitution

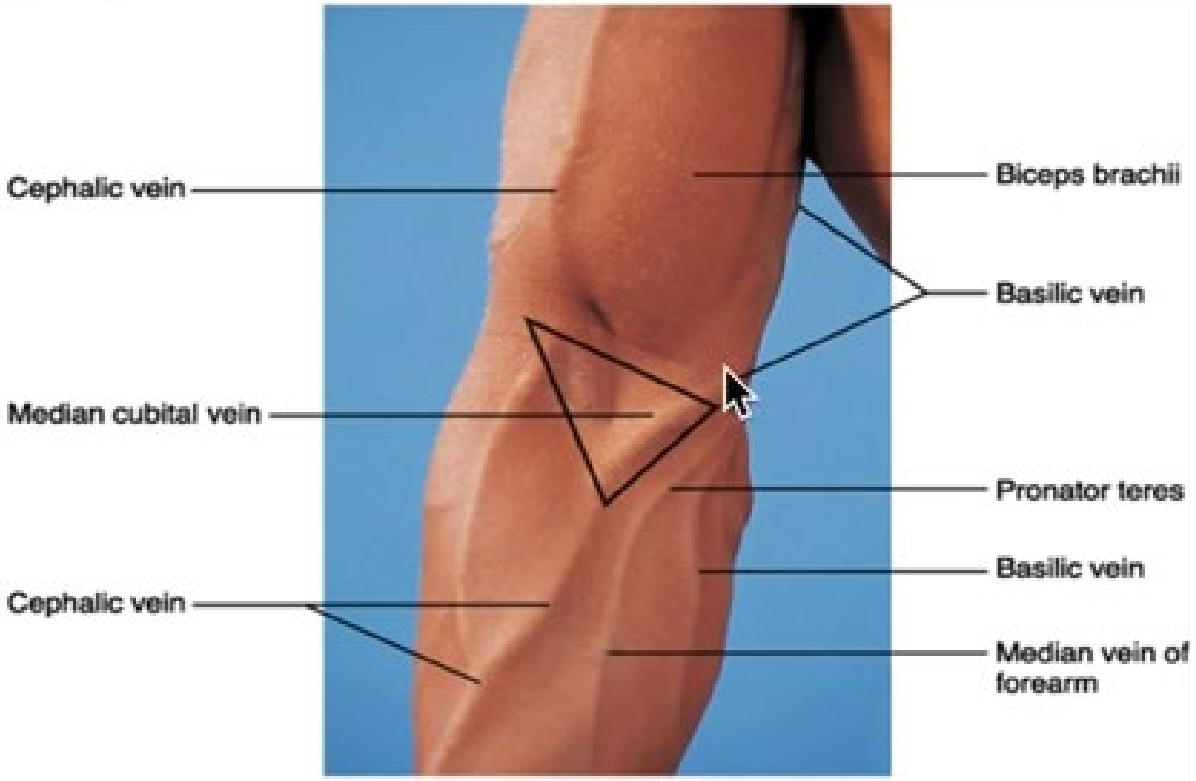


Vascular Access

- Size & Placement
- Flow Rates
  - Poiseuille's Law
- Maintenance
  - Dressing Changes
  - Saline Lock
  - Flushes
- Labelling & site rotation
- Infiltration/Extravasation

Vessel Selection





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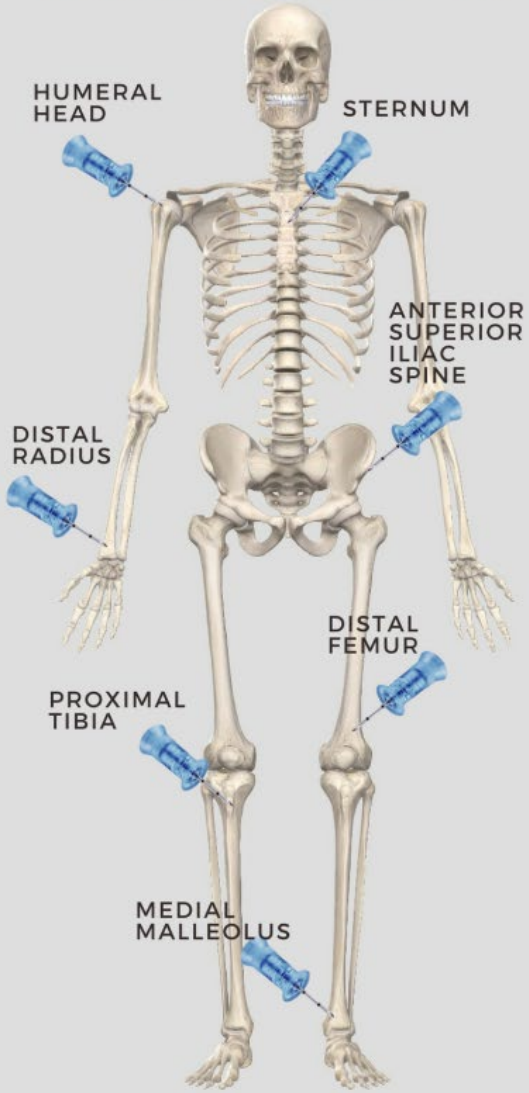
iEM
INTERNATIONAL EMERGENCY MEDICINE EDUCATION PROJECT

INTRAOSSEOUS LINES AND ACCESS SITES



YELLOW - (45 MM) FOR LARGE PATIENTS OR DENSE BONE SITES

BLUE - (25 MM) FOR PATIENTS 40 KG AND GREATER

PINK - (15 MM) FOR PATIENTS 3 – 39 KG

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IV site Maintenance

- If inserted under non-aseptic conditions, replace vascular access device (VAD) within 24-48 hours
- Flush line with syringe of preservative-free 0.9% Normal Saline (NS).
- Change dressing and cleanse site q7 days or more often if integrity of dressing is compromised or complications arise.
- Re-site only if signs of complications and VAD is still needed for therapy.
- Scrub connections for 15 seconds prior to use

IV Line Maintenance

- Label all administration sets with date and time
- Label IV-line close to patient with medication

Change IV tubing:

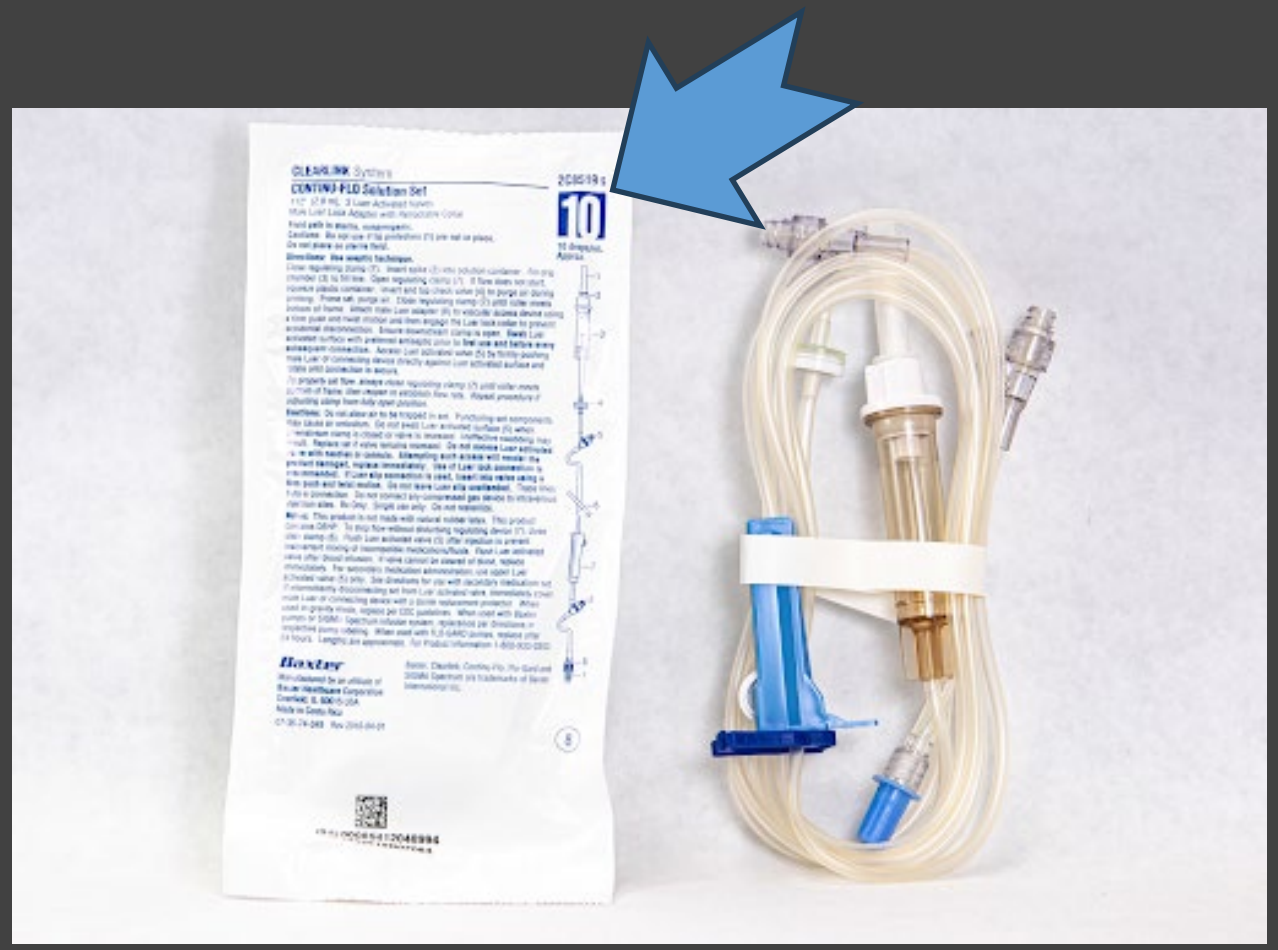
- With the insertion of a new VAD.
- When the IV solution has changed.
- Immediately if contamination or integrity of the system/product has occurred.
- Otherwise change set as per table:

Administration	Frequency of Administration Set Change
Uninterrupted continuous infusion	q7days
Intermittent infusions (includes both primary and secondary sets)	q24h
Parenteral Nutrition	q24h and with bag changes or disconnections (as per Parenteral Nutrition policy)
Lipids	q12h
Propofol	q6-12h
Blood and blood products	After 4 units or 4 hours whichever comes first. Between different products.

Fluid Administration

Method of Administration:

1. By Gravity
   - *must calculate drip rate for tubing*
2. Pressure Bags
   - *be mindful of flow on IV gauges
3. IV Pump
   - *allows for specific volume to be delivered



Drip Factor

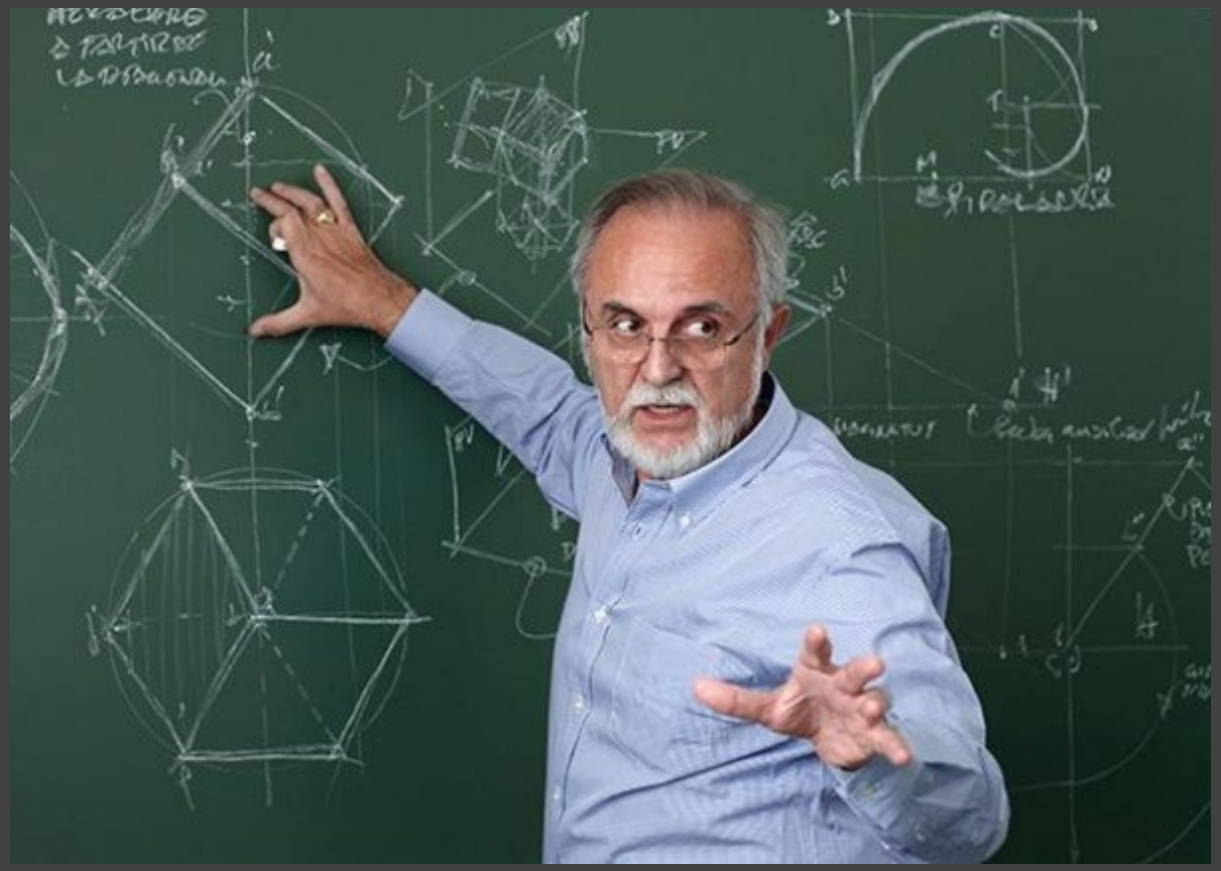
Convert ml/hr to gtts/min

Calculation:

$$\frac{(\text{ml/hr}) \times \text{Drop Factor}}{60}$$

Practice:

What is the drip rate you would need with this IV tubing to achieve 100ml/hr?



Solution:

100ml/hr:

$$= \frac{100 \times 10}{60}$$

$$= \frac{1000}{60}$$

$$= 16.66666$$

How to apply:

Drip Rate of 17gtts/min

Now What?

Convert the gtts/min to a measurable number by dividing by 4
*this will provide the number of drops in 15 seconds

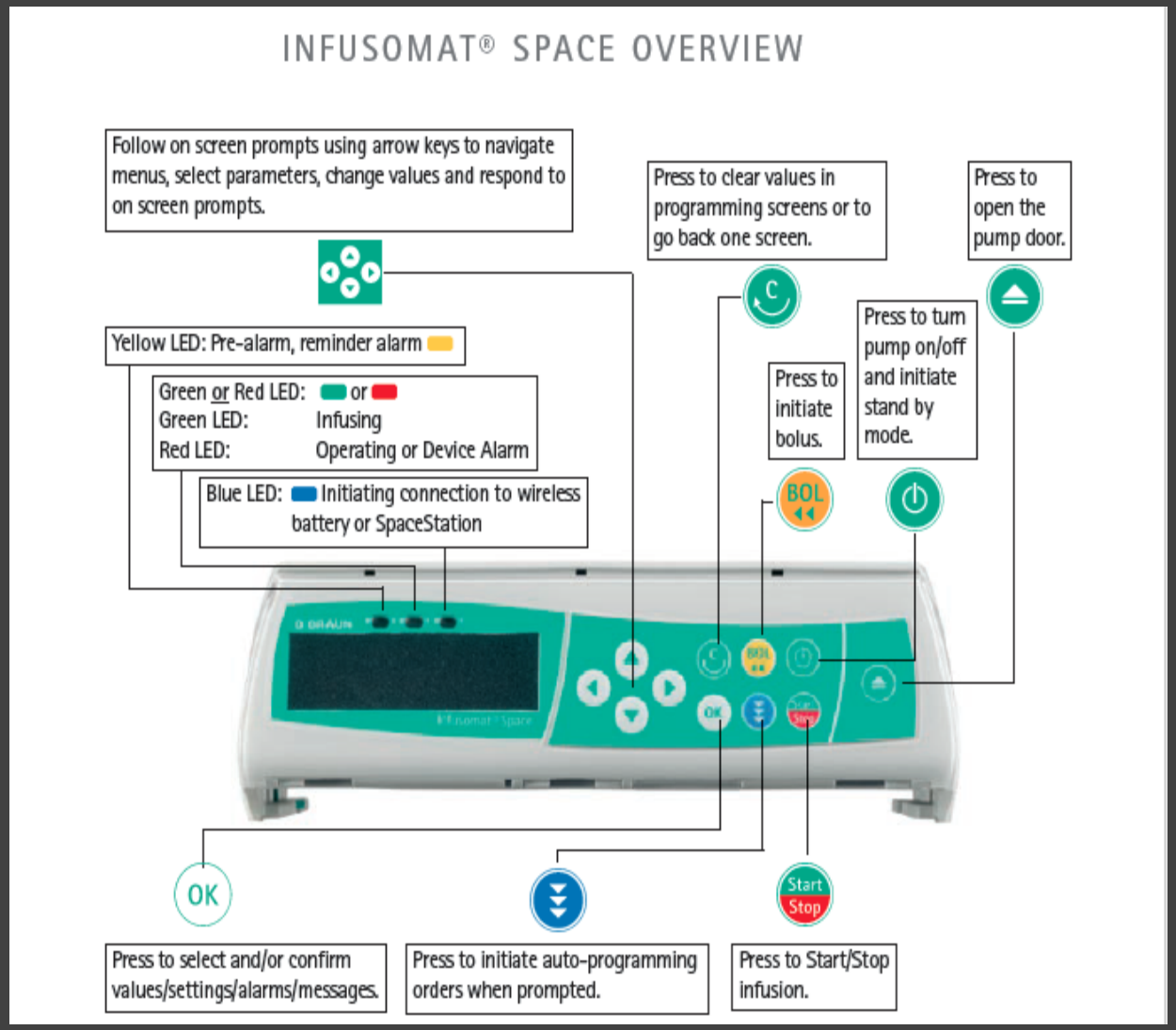
100ml/hrX10drop factor / 60 (minutes) = 17gtt/min
divide by 4 = **4 drops/15seconds**

Pressure Bag Infusion

When do you need to use them?

What if you don't have one?

*remember flowrates for IV gauge



IV Pump Infusion

BBraun Infusomat Space Pump

- Programmable for primary, secondary, and basic infusions
- Drug Infusion Library
  - Critical Care contains Inotropes and vasopressors
  - Generic contains 1200 drug names and concentrations
    *Be aware of different concentrations

Programming the Pump

Turn On Pump

- Press & hold “on/off” button

Install Tubing

- Press and hold door open
- Clamp line confirmation
- Install tubing from right to left, only one way for white piece (O & D)
- Be gentle- can break, still works, not ideal
- Press green end in, hook out until green press button pops out
- Close door with two fingers, hold
- Wait to open clamp

Priming

- Either prime manually by taking anti gravity cap off
- OR prime by pump

Buttons

- Can use okay button or left arrow button to select on menu
- C/back button
- Any button silences alarms

Programming the Pump

Program a Basic Infusion

Program two of the following:

- Rate
- VTBI
- Time

Program a Secondary Infusion

- Primary Infusion must be stopped
- Scroll to Secondary Infusion
- Select “No” to “Use Dose Rate Calculation”
- Program two of the following
  - Rate
  - VTBI
  - Time

    Recommendation: VTBI & Time, Rate is calculated for you

- Select Start
- Upon completion the pump will automatically return to the primary infusion

Programming the Pump

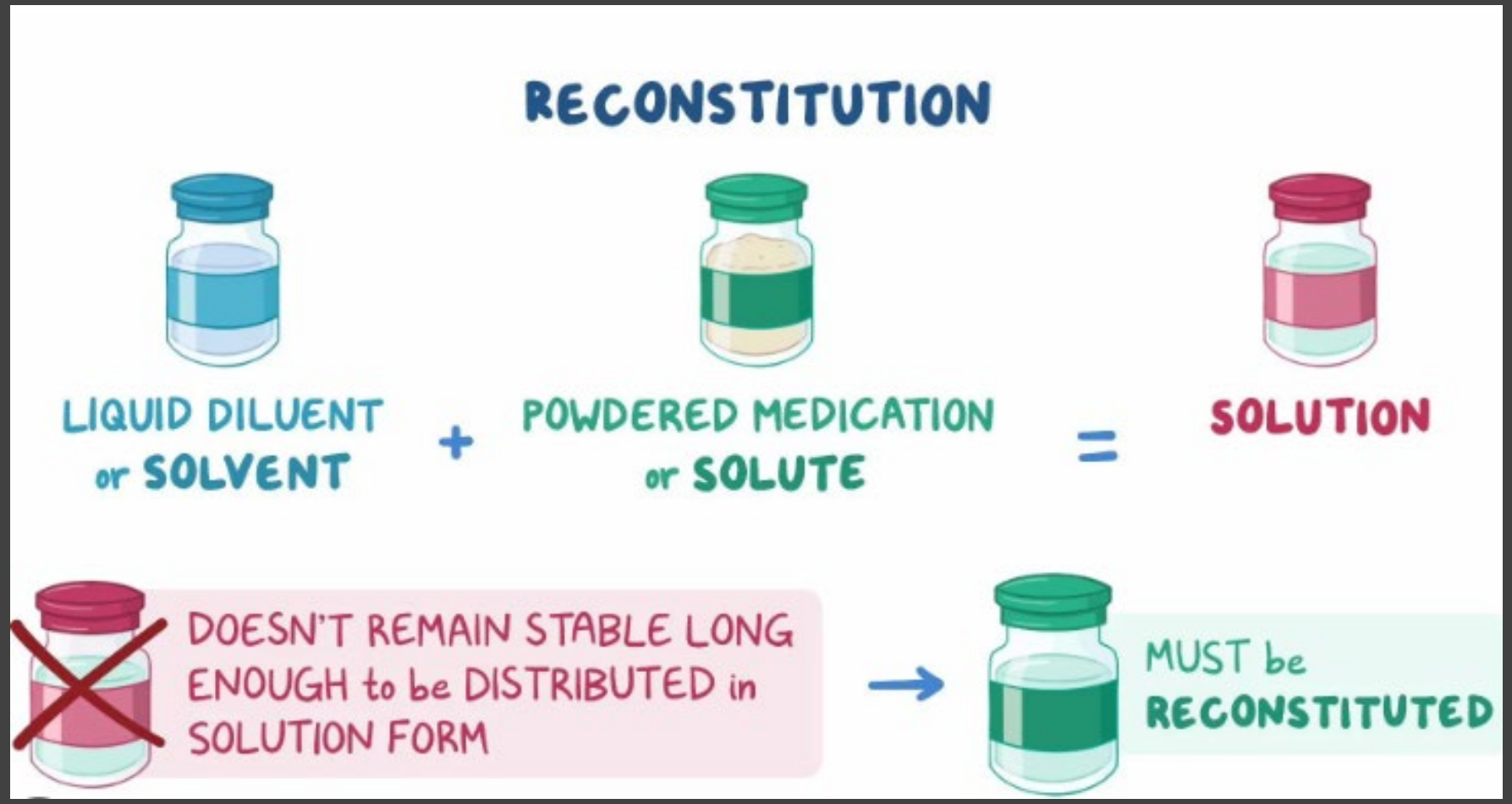
Enter Standby

- Press and hold on/off button
- Will save settings until runs out of batteries or reprogrammed
- C cancels standby

Remove Tubing

- Stop infusion
- Press ‘eject’ button
- Follow directions on screen
- May need to pull to detatch green hook

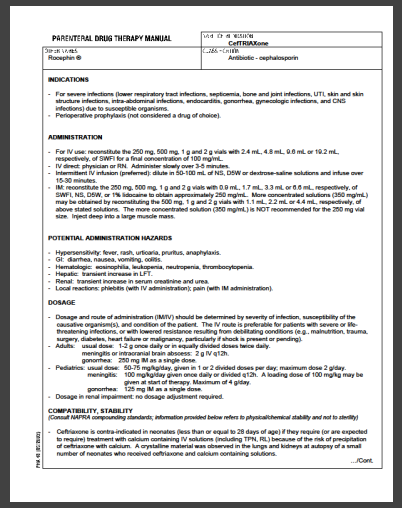
Medication Reconstitution



Reconstitution

DRUG MONOGRAPH INFO

1. Indications
2. Reconstitution and Stability
3. Compatibility
4. Preparation and Administration
5. Dosages
6. Complications



PARENTERAL DRUG THERAPY MANUAL

Ceftriaxone

Rocephin ®

Antibiotic - cephalosporin

INDICATIONS

- For severe infections (lower respiratory tract infections, septicemia, bone and joint infections, UTI, skin and skin structure infections, intra-abdominal infections, endocarditis, gonorrhea, gynecologic infections, and CNS infections) due to susceptible organisms.
- Perioperative prophylaxis (not considered a drug of choice).

ADMINISTRATION

- For IV use: reconstitute the 250 mg, 500 mg, 1 g and 2 g vials with 2.4 mL, 4.8 mL, 9.6 mL or 19.2 mL, respectively, of SWFI for a final concentration of 100 mg/mL.
- IV direct: physician or RN. Administer slowly over 3-5 minutes.
- Intermittent IV infusion (preferred): dilute in 50-100 mL of NS, D5W or dextrose-saline solutions and infuse over 15-30 minutes.
- IM: reconstitute the 250 mg, 500 mg, 1 g and 2 g vials with 0.9 mL, 1.7 mL, 3.3 mL or 6.6 mL, respectively, of SWFI, NS, D5W, or 1% lidocaine to obtain approximately 250 mg/mL. More concentrated solutions (350 mg/mL) may be obtained by reconstituting the 500 mg, 1 g and 2 g vials with 1.1 mL, 2.2 mL or 4.4 mL, respectively, of above stated solutions. The more concentrated solution (350 mg/mL) is NOT recommended for the 250 mg vial size. Inject deep into a large muscle mass.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: fever, rash, urticaria, pruritus, anaphylaxis.
- GI: diarrhea, nausea, vomiting, colitis.
- Hematologic: eosinophilia, leukopenia, neutropenia, thrombocytopenia.
- Hepatic: transient increase in LFT.
- Renal: transient increase in serum creatinine and urea.
- Local reactions: phlebitis (with IV administration); pain (with IM administration).

DOSAGE

- Dosage and route of administration (IM/IV) should be determined by severity of infection, susceptibility of the causative organism(s), and condition of the patient. The IV route is preferable for patients with severe or life-threatening infections, or with lowered resistance resulting from debilitating conditions (e.g., malnutrition, trauma, surgery, diabetes, heart failure or malignancy, particularly if shock is present or pending).
- Adults: usual dose: 1-2 g once daily or in equally divided doses twice daily.
  meningitis or intracranial brain abscess: 2 g IV q12h.
  gonorrhea: 250 mg IM as a single dose.
- Pediatrics: usual dose: 50-75 mg/kg/day, given in 1 or 2 divided doses per day; maximum dose 2 g/day.
  meningitis: 100 mg/kg/day given once daily or divided q12h. A loading dose of 100 mg/kg may be given at start of therapy. Maximum of 4 g/day.
  gonorrhea: 125 mg IM as a single dose.
- Dosage in renal impairment: no dosage adjustment required.

COMPATIBILITY, STABILITY

(Consult MATRIX compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Ceftriaxone is contra-indicated in neonates (less than or equal to 28 days of age) if they require (or are expected to require) treatment with calcium containing IV solutions (including TPN, RL) because of the risk of precipitation of ceftriaxone with calcium. A crystalline material was observed in the lungs and kidneys at autopsy of a small number of neonates who received ceftriaxone and calcium containing solutions.

.../Cont.

PM 46 (02/2003)

Ceftriaxone

Reconstitution and Stability

For IV administration reconstitute with sterile water for injection (SWI) as follows:

- 1 g vial: add 9.6 mL to provide 100 mg/mL solution
- 2 g vial: add 19.2 mL to provide 100 mg/mL solution

Intermittent IV : preferred route

- dilute in 50 mL;
- administer over 20 to 30 minutes

Practice

- Apply tourniquet to a partner, look for PIV locations
- Prime a gravity infusion and set drip for 100ml/hr
- Add a secondary infusion of 1 g Ceftriaxone to primary drip infusion (155)
- Start a primary infusion on the pump of NS at 150 ml/hr
- Add a secondary infusion on the pump of Pantoloc 80 mg (511)
- Stop the infusion, put the pump on standby
- Remove the secondary, change the primary infusion to propofol at 20 mcg/kg/min (page 556)
- Bolus 10 mg on the pump then change the rate to 30 mcg/kg/min.

References

Drug Monographs – the Ottawa, PEPID, 1 Cdn Fd Hosp

http://pdtm.vch.ca/monograph-information/monographs#group-p

Personal Cheat Sheets

EMCRit

LITFL

CrackCast

Braun Manual/You Tube Videos